

**B.Tech. Degree VI Semester Regular/Supplementary Examination
May 2024****CE 19-201-0601 ENVIRONMENTAL ENGINEERING-I**
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Recognize the important professional and ethical responsibilities as an environmental engineer so as to estimate or analyze the quantity and quality of water required for a community water supply scheme.
- CO2: Apply perfect knowledge on water supply sources for the design of collection, transport, transmission systems.
- CO3: Demonstrate knowledge on sanitary plumbing systems, design of sewerage systems and distribution systems in water supply engineering.
- CO4: Recognize various natural methods of waste water disposal and self purification of streams.
- CO5: Demonstrate an ability to provide engineering solutions for the environmental problems related with air pollution solid waste disposal and noise pollution.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A
(Answer ALL questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
1. (a)	What is meant by Ideal Population Growth curve?	3	3	L1	1	3,6
(b)	Define fire demand. List out any two formula to calculate fire demand.	3	3	L2	1	1,2
(c)	What is meant by intake tower?	3	3	L2	2	3,6
(d)	Which are the various classification of pumps?	3	3	L2	2	3,6
(e)	Which are the different shapes of sewers?	3	3	L1	3	3,6
(f)	Explain the important features of sewer junction.	3	3	L3	3	3,6
(g)	What is meant by critical Dissolved oxygen deficit?	3	3	L2	4	3,6
(h)	What are the various effects of air pollution in plants?	3	3	L1	5	3,6

PART B

(4 × 12 = 48)

- II. Determine the future population of a satellite town by the Geometric Increase method for the year 2021, given the following data:

Year	1951	1961	1971	1981	1991	2021
Population	70,000	1,00,000	1,50,000	2,00,000	2,40,000	?

OR

- III. In two periods each of 20 years, a city has grown from 50,000 to 1,70,000 and then to 3,00,000. Determine (i) the saturation population (ii) the equation of the logistic curve (iii) the expected population after the next 20 years.

- IV. What are the various factors to be considered in the location of an intake? Explain in detail with neat sketches the various types of intakes.

OR

(P.T.O)

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<u>V.</u>	Explain in detail about the various types of distribution system.	Marks 12	BL L2, L4	CO 2	PO 3,6
VI.	Describe in detail about the maintenance, inspection and ventilation of sewers.	12	L2, L4	3	1,3
OR					
<u>VII.</u>	Explain in detail about the different materials used in sewer constructions.	12	L2, L5	3	3,6
<u>VIII.</u>	Draw the oxygen sag curve and various zones of pollution and explain about the self-purification phenomena in rivers.	12	L2, L5	4	1,3
OR					
IX.	What are the different types of air pollutants? Also relate the air pollution effects on human health and meteorological aspects.	12	L2, L5	5	1,3

Bloom's Taxonomy Levels

L1 - 15%, L2 - 60%, L3 - 5%, L4 - 10%, L5 - 10%.
